

14.8 A 100-mm-long, hollow iron cylinder is exposed to a 1000°C carburizing gas (a mixture of CO and CO₂) at its inner and outer surfaces of radii 4.30 and 5.70 mm, respectively. Consider steady-state conditions for which carbon diffuses from the inner surface of the iron wall to the outer surface and the total transport amounts to 3.6×10^{-3} kg of carbon over 100 h. The variation of the carbon composition (weight % carbon) with radius is tabulated for selected radii.

r (mm)	4.49	4.66	4.79	4.91	5.16	5.27	5.40	5.53
Wt.C (%)	1.42	1.32	1.20	1.09	0.82	0.65	0.46	0.28

(a) Beginning with Fick's law and the assumption of a constant diffusion coefficient, D_{C-Fe} , show that $dp_C/d(\ln r)$ is a constant. Sketch the carbon mass density, $\rho_C(r)$, as a function of $\ln r$ for such a diffusion process.

(b) The foregoing table corresponds to measured distributions of the carbon mass density. Is D_{C-Fe} constant for this diffusion process? If not, does D_{C-Fe} increase or decrease with an increasing carbon concentration?

(c) Using the experimental data, calculate and tabulate D_{C-Fe} for selected carbon compositions.

a) Fick's law for cylindrical coordinates

$$j_c = -D_{Fe} A_r \frac{d\rho_c}{dr}$$

$$j_c = -D_{Fe} (2\pi L) \frac{d\rho_c}{dr}$$

We have steady state system

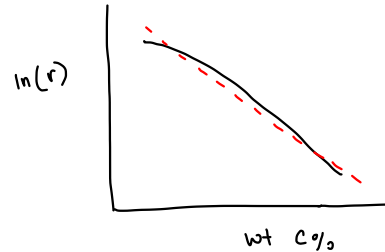
$$j_c = \text{constant}$$

$$-D_{Fe} (2\pi L) \frac{d\rho_c}{dr} = C$$

$$r \frac{d\rho_c}{dr} = C$$

$$\boxed{\frac{d\rho_c}{d(\ln r)} = C}$$

$\ln(r)$	Wt.C %
1.502	1.42
1.539	1.32
1.567	1.20
1.591	1.09
1.641	0.82
1.662	0.65
1.686	0.46
1.71	0.28



plot is done on excel sheet

b) D_{Fe} is not a constant because it doesn't fit on to the linear line

$$\therefore D_{Fe} \frac{d\rho_c}{d(\ln(r))} = C$$

D_{Fe} and C are proportional $\rightarrow D_{Fe}$ increases when C increases.

$$c) j_c = \frac{3.6 \times 10^{-3} \text{ kg}}{100 \text{ h}} \times \frac{3600 \text{ s}}{1 \text{ h}} = 1 \times 10^{-8} \text{ kg/s}$$

$$\text{When } r = 4.66 \quad \text{Wt C\%} = 1.32$$

$$\frac{d \text{Wt C\%}}{dr} = \frac{1.20 - 1.42}{4.79 - 4.49} = -0.73$$

$$j_c = -\rho D_{Fe} A_r \frac{d(\text{Wt C\%})}{dr}$$

$$(1 \times 10^{-8}) = -(7730) (D_{Fe}) \left(2\pi \left(\frac{4.66}{1000} \right) \left(\frac{100}{1000} \right) (-0.73) \right)$$

$$\boxed{D_{Fe} = 6.05 \times 10^{-10} \text{ m}^2/\text{s}} \quad \text{x too high}$$

$$\text{When } r = 5.40 \quad \text{Wt C\%} = 0.46$$

$$\frac{d \text{Wt C\%}}{dr} = \frac{0.28 - 0.65}{5.53 - 5.27} = -1.42$$

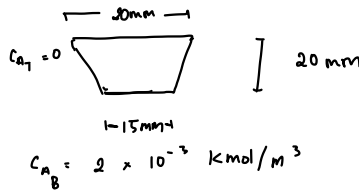
$$j_c = -\rho D_{Fe} A_r \frac{d(\text{Wt C\%})}{dr}$$

$$(1 \times 10^{-8}) = -(7730) (D_{Fe}) \left(2\pi \left(\frac{5.40}{1000} \right) \left(\frac{100}{1000} \right) (-1.42) \right)$$

$$\boxed{D_{Fe} = 2.69 \times 10^{-10} \text{ m}^2/\text{s}} \quad \text{x too high}$$

14.10 An old-fashioned glass apothecary jar contains a patent medicine. The neck is closed with a rubber stopper that is 20 mm tall, with a diameter of 15 mm at the bottom end, widening to 20 mm at the top end. The molar concentration of medicine vapor in the stopper is $2 \times 10^{-3} \text{ kmol/m}^3$ at the bottom surface and is negligible at the top surface. If the mass diffusivity of medicine vapor in rubber is $0.15 \times 10^{-9} \text{ m}^2/\text{s}$, find the rate (kmol/s) at which vapor exits through the stopper.

(33/33)



$$D_{AB} = 0.15 \times 10^{-9} \text{ m}^2/\text{s}$$

- Assumptions
- 1) 1 dimensional
 - 2) steady state
 - 3) Constant properties
 - 4) No reactions

Fick's Law of Molar

$$J_A^* = -C D_{AB} \nabla x_A = -C D_{AB} \frac{dx_A}{dx} = -D_{AB} \frac{dC_A}{dx} \quad (\text{eq. 14.13})$$

$$N_A = J_A^* A_c = -D_{AB} A_c \frac{dC_A}{dx}$$

$$\int_0^L N_A \frac{dx}{A_c} = \int_{x_B}^{x_T} -D_{AB} dC_A \Rightarrow N_A \int_0^L \frac{dx}{A_c} = -D_{AB} \int_{x_B}^{x_T} dC_A = -D_{AB} (C_{A1} - C_{AB}) = D_{AB} C_{AB}$$

neglectable

$$A_c = \pi R^2$$

$$R = \frac{R_1 + (R_2 - R_1)x}{L}$$

$$\Rightarrow \frac{L dR}{R_2 - R_1} = dx$$

$$N_A \int_{R_1}^{R_2} \frac{L dR}{(\pi R^2)(R_2 - R_1)} = D_{AB} C_{AB}$$

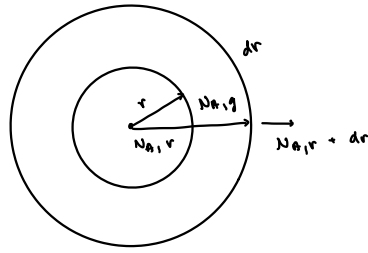
$$\frac{N_A}{\pi} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \left(\frac{L}{R_2 - R_1} \right) = D_{AB} C_{AB}$$

$$N_A = \frac{D_{AB} C_{AB} \pi}{\left(\frac{1}{R_1} - \frac{1}{R_2} \right) \left(\frac{L}{R_2 - R_1} \right)} = \frac{(0.15 \times 10^{-9}) (2 \times 10^{-3}) (\pi)}{\left(\frac{1}{0.0075} - \frac{1}{0.01} \right) \left(\frac{0.02}{0.01 - 0.0075} \right)} =$$

$$N_A = 3.53 \times 10^{-15} \text{ kmol/s}$$

14.19 Beginning with a differential control volume, derive the diffusion equation, on a molar basis, for species A in a one-dimensional, spherical, stationary medium, considering species generation. Compare your result with [Equation 14.50](#).

(33/33)



Conservation equation for molar

$$N_{A,r} + N_{A,\dot{q}} - N_{A,r+dr} = \frac{dN_A}{dt}$$

$$N_{A,r} = -D_{AB} 4\pi r^2 \frac{\partial C_A}{\partial r}$$

$$N_{A,r+dr} = N_{A,r} + \frac{\partial N_{A,r}}{\partial r} dr$$

$$N_{A,\dot{q}} = N_A (4\pi r^2 dr)$$

$$\frac{dN_A}{dt} = \frac{\partial C_A 4\pi r^2 dr}{\partial t}$$

$$-D_{AB} 4\pi r^2 \frac{\partial C_A}{\partial r} + N_A (4\pi r^2 dr) - \left(N_{A,r} + \frac{\partial N_{A,r}}{\partial r} dr \right) = \frac{\partial C_A 4\pi r^2 dr}{\partial t}$$

$$-D_{AB} 4\pi r^2 \frac{\partial C_A}{\partial r} + N_A (4\pi r^2 dr) - N_{A,r} - \frac{\partial N_{A,r}}{\partial r} dr = \frac{\partial C_A 4\pi r^2 dr}{\partial t}$$

$$N_A (4\pi r^2 dr) + 4\pi r^2 \frac{\partial}{\partial r} \left(D_{AB} r^2 \frac{\partial C_A}{\partial r} \right) dr = 4\pi r^2 \frac{\partial C_A}{\partial t} dr$$

$$\boxed{N_A + \frac{1}{r^2} \frac{\partial}{\partial t} \left(D_{AB} r^2 \frac{\partial C_A}{\partial r} \right) = \frac{\partial C_A}{\partial t}}$$

Eq 14.50 $\frac{1}{r^2} \frac{\partial}{\partial t} \left(C D_{AB} r^2 \frac{\partial x_A}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \phi} \left(C D_{AB} \frac{\partial x_A}{\partial \phi} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(C D_{AB} \sin \theta \frac{\partial x_A}{\partial \theta} \right) + \dot{N}_A = \frac{\partial C_A}{\partial t}$

We can reduce eq. 14.50 to the eq. we found if C is independent of r and θ + ϕ are neglectable